

UNIVERSITA' DEGLI STUDI DI ROMA TOR VERGATA

ESERCITAZIONE CORSO DI ANALISI MATEMATICA I

CORSO DI LAUREA IN INGEGNERIA

ESERCITATORE: DANIELE PASQUAZI

pasquazi@mat.uniroma2.it

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1. Verificare che:

$$x^2 = o(x^3) \quad \text{per } x \rightarrow +\infty$$

$$x^2 = o(2^x) \quad \text{per } x \rightarrow +\infty$$

$$x^3 = o(x^2) \quad \text{per } x \rightarrow 0$$

$$(\log_3 x)^2 = o\left(x^{\frac{1}{100}}\right) \quad \text{per } x \rightarrow +\infty$$

2. Verificare le seguenti uguaglianze:

$$1 - \cos(2x) + x^2(1 + \log(1 + x)) = 3x^2 + o(x^2) \\ \text{per } x \rightarrow 0^+$$

$$e^{\frac{1}{x}} = o(x^n) \\ \text{per } x \rightarrow 0^- \quad \forall n > 0$$

$$e^{-3/\sqrt{x}} = 1 - 3/\sqrt{x} + o(1/\sqrt{x}) \\ \text{per } x \rightarrow +\infty$$

$$\log x = o(x^{-\alpha}) \\ \text{per } x \rightarrow 0^- \quad \forall \alpha > 0$$

3. Calcolare i seguenti limiti

$$3.a \quad \lim_{x \rightarrow 0^+} \frac{\sqrt{1+x} - 1}{x(x^2 + 1)}$$

$$3.m \quad \lim_{x \rightarrow +\infty} \frac{\log(1 + x^2) + x}{x(1 + x^2)}$$

$$3.b \quad \lim_{x \rightarrow +\infty} \frac{4x^3 + x^2}{2x^5 + 2x}$$

$$3.n \quad \lim_{x \rightarrow 0^+} \frac{\log(1 + x^2) + x \log x}{x(1 + \log x)}$$

$$3.c \quad \lim_{x \rightarrow 0} \frac{4x^3 + x^2}{2x^5 + 2x}$$

$$3.o \quad \lim_{x \rightarrow +\infty} \frac{\log(1 + x^2) + x \log x}{x(1 + \log x)}$$

$$3.d \quad \lim_{x \rightarrow -3} \frac{x^3 - x^2 - 8x + 12}{x^2 + 2x - 3}$$

$$3.p \quad \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\log(x^2 + 1)}$$

$$3.e \quad \lim_{x \rightarrow \infty} \left(\frac{x+1}{\sqrt{x}} - \frac{x}{\sqrt{x+1}} \right)$$

$$3.q \quad \lim_{x \rightarrow 0^+} \frac{\log(1-x)}{\sqrt{1+x} - e^x}$$

$$3.f \quad \lim_{x \rightarrow 0} \frac{1 - \cos(x^2)}{x^4}$$

$$3.r \quad \lim_{x \rightarrow 0} (\cos^2 x)^{1/x^2}$$

$$3.g \quad \lim_{x \rightarrow 0^+} \frac{\log(1 + x^2)}{(e^x - 1)^2}$$

$$3.s \quad \lim_{x \rightarrow 0} (1 - \sin^2 x)^{\frac{-1}{\sin^2 x}}$$

$$3.h \quad \lim_{x \rightarrow 0^+} \frac{e^{2x \sin(3x)} - 1}{1 - \cos x}$$

$$3.t \quad \lim_{x \rightarrow 0} \frac{\log(1 + x^2) \sqrt{e^x - 1}}{(1 - \cos x) \sqrt{\sin(x)}}$$

$$3.i \quad \lim_{x \rightarrow 0} \frac{\sqrt[3]{x \sin^2(2x)}(\sqrt{x} - 1)}{e^x - \cos(3x)}$$

$$3.u \quad \lim_{x \rightarrow 0} \frac{\sin(e^x - e^{-x})}{(1 - \cos x)}$$

$$3.l \quad \lim_{x \rightarrow 0^+} \frac{\log(1 + x^2) + x}{x(1 + x^2)}$$

$$3.v \quad \lim_{x \rightarrow 0^+} \log x \cdot \log \left(1 + \frac{1}{x}\right)$$

$$3.w \quad \lim_{x \rightarrow 1^+} \frac{e^{x^2-1}}{\sqrt{\log x}}$$

$$3.x \quad \lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)^3}{\sin(x - 1)}$$

Soluzioni

(3. a) $1/2$;

(3. b) 0 ;

(3. c) 0 ;

(3. d) $-25/4$;

(3. e) 0 ;

(3. f) $1/2$;

(3. g) 1 ;

(3. h) 12 ; $\frac{e^{2x \sin(3x)} - 1}{1 - \cos x} = \frac{2x \sin(3x)}{\frac{1}{2}x^2} (1 + o(1)) = \frac{4 \cdot 3x}{x} (1 + o(1)) = 12(1 + o(1)) \xrightarrow{x \rightarrow 0^+} 12$

(3. i) $-\sqrt[3]{4}$; $\frac{\sqrt[3]{x \sin^2(2x)}(\sqrt{x} - 1)}{e^x - \cos(3x)} = \frac{x(\sqrt[3]{4} + o(1))(o(1) - 1)}{x(1 + 9x/2 + o(1))} \xrightarrow{x \rightarrow 0} -\sqrt[3]{4}$

(3. l) 1 ;

(3. m) 0 ;

(3. n) 1 ;

(3. o) 1 ;

(3. p) non esiste;

(3. q) 2 ;

(3. r) e^{-1} ;

(3. s) e ;

(3. t) 2 ; $\frac{\log(1+x^2)\sqrt{e^x-1}}{(1-\cos x)\sqrt{\sin(x)}} = \frac{x^2(1+o(1))\sqrt{x(1+o(1))}}{\frac{1}{2}x^2(1+o(1))\sqrt{x(1+o(1))}} \xrightarrow{x \rightarrow 0} 2$

(3. u) non esiste;

(3. v) $-\infty$.

(3. w) 0

(3. x) 0